

```
library(car)
library(graphics)
library (sm)
library(BSDA)
library(tseries)
library(lmtest)
library(MASS)
library(nortest)
library(e1071)
library(SiZer)
library(exactRankTests)
library(lmPerm)
library(effsize)
```

```
#####
#####
#####
#####
##### data analysis: controlled experiment
```

```
## caricare il dataset
```

```
dati <- read.table("/Users/carminc/Documents/CORSI/Metriche e
qualità del software/LEZIONI/L9 data experiment.csv", header =
TRUE, sep = ";", dec = ".")
```

```
##### selection of distributions
```

```
CD <- dati[which(dati["METHOD"] == "CD"),]
ER <- dati[which(dati["METHOD"] == "ER"),]
```

```
#### descriptive statistics
```

```
summary(CD$CS)
summary(ER$CS)
```

```
### standard deviation
```

```
sd(CD$CS)
sd(ER$CS)
```

```
# boxplot
```

```

boxplot(CD$CS, ER$CS,
#range = 1.5,
#xlim = c(0.5, 3.5),
#ylim = c(0, 35)
boxwex = 0.25,
names=c("CD", "ER"),
main="boxplot of F-measure for CD and ER",
#xlab="(a)",
outline = TRUE, col = "gray")

#### interaction plot of FM considering TREAT and TASK
par(cex=1)
interaction.plot(dati$METHOD, dati$ABILITY, dati$CS, trace.label =
"ABILITY",
xlab = "METHOD", ylab = "Mean of F-measure", ylim = c(0, 1),
cex.axis=1.5, cex.lab=1.5, cex.main=1.5)

an <- anova(lm(FM ~ METHOD*SYSTEM*LAB, dati))
an

##### statistical significance difference

t.test(CD$CS, ER$CS, alternative="two.side", paired = TRUE,
conf.level = 0.95)

#####

wilcox.test(CD$CS, ER$CS, alternative="two.side", paired = TRUE,
conf.level = 0.95)
### paired=FALSE for Mann-Whitney test

##### effect size

VD.A(CD$CS, ER$CS)

cliff.delta(CD$CS, ER$CS)

```


#####

Two-sample Mann–Whitney U Test

The two-sample Mann–Whitney U test is a rank-based test that compares values for two groups. A significant result suggests that the values for the two groups are different. It is equivalent to a two-sample Wilcoxon rank-sum test.

Without further assumptions about the distribution of the data, the Mann–Whitney test does not address hypotheses about the medians of the groups. Instead, the test addresses if it is likely that an observation in one group is greater than an observation in the other. This is sometimes stated as testing if one sample has stochastic dominance compared with the other.

The test assumes that the observations are independent. That is, it is not appropriate for paired observations or repeated measures data.

The test is performed with the *wilcox.test* function.

Appropriate effect size statistics include Vargha and Delaney's *A*, Cliff's *delta*, among others.

Appropriate data

- Two-sample data. That is, one-way data with two groups only
- Dependent variable is ordinal, interval, or ratio
- Independent variable is a factor with two levels. That is, two groups
- Observations between groups are independent. That is, not paired or repeated measures data
- In order to be a test of medians, the distributions of values for each group need to be of similar shape and spread. Otherwise the test is typically a test of stochastic equality.

Hypotheses

- Null hypothesis: The two groups are sampled from populations with identical distributions. Typically, that the sampled populations exhibit stochastic equality.
- Alternative hypothesis (two-sided): The two groups are sampled from populations with different distributions. Typically, that one sampled population exhibits stochastic dominance.

Interpretation

Significant results can be reported as e.g. “Values for group A were significantly different from those for group B.”

Other notes and alternative tests

The Mann–Whitney U test can be considered equivalent to the Kruskal–Wallis test with only two groups.

Mood's median test compares the medians of two groups. It is described in its own chapter.

For ordinal data, an alternative is to use cumulative link models, which are described later in this book.

Effect size

Statistics of effect size for the Mann–Whitney test report the degree to which one group has data with higher ranks than the other group. They are related to the probability that a value from one group will be greater than a value from the other group. Unlike p -values, they are not affected by sample size.

Vargha and Delaney's A is relatively easy to understand. It reports the probability that a value from one group will be greater than a value from the other group. A value of 0.50 indicates that the two groups are stochastically equal. A value of 1 indicates that the first group shows complete stochastic domination over the other group, and a value of 0 indicates the complete stochastic domination by the second group.

Cliff's δ is linearly related to Vargha and Delaney's A . It ranges from -1 to 1 , with 0 indicating stochastic equality of the two groups. 1 indicates that one group shows complete stochastic domination over the other group, and a value of -1 indicates the complete stochastic domination of the other group. Its absolute value will be numerically equal to Freeman's θ .

A common effect size statistic for the Mann–Whitney test is r , which is the Z value from the test divided by the total number of observations. As written here, r varies from 0 to close to 1 . In some formulations, it varies from -1 to 1 .

Kendall's τ -b is sometimes used, and varies from approximately -1 to 1 .

Freeman's θ and ϵ -squared are usually used when there are more than two groups, with the Kruskal–Wallis test, but can also be employed in the case of two groups.

Interpretation of effect sizes necessarily varies by discipline and the expectations of the experiment, but for behavioral studies, the guidelines proposed by Cohen (1988) are sometimes followed. The following guidelines are based on the literature values and my personal intuition. They should not be considered universal.

Optional technical note: The interpretation values for r below are found commonly in published literature and on the internet. I suspect that this interpretation stems from the adoption of Cohen's interpretation of values for Pearson's r . This may not be justified, but it turns out that this interpretation for the r used here is relatively reasonable. The interpretation for τ -b, Freeman's θ , and ϵ -squared here are based on their values relative to those for r , based on simulated data (5-point Likert items, n per group between 4 and 25). Plots for some of these simulations are shown below.

Interpretations for Vargha and Delaney's A and Cliff's δ come from Vargha and Delaney (2000).

	<u>small</u>	<u>medium</u>	<u>large</u>
<i>r</i>	0.10 – < 0.30	0.30 – < 0.50	≥ 0.50
<i>tau-b</i>	0.10 – < 0.30	0.30 – < 0.50	≥ 0.50
Cliff's delta	0.11 – < 0.28	0.28 – < 0.43	≥ 0.43
Vargha and Delaney's A	0.56 – < 0.64	0.64 – < 0.71	≥ 0.71
	> 0.34 – 0.44	> 0.29 – 0.34	≤ 0.29
Freeman's theta	0.11 – < 0.34	0.34 – < 0.58	≥ 0.58
epsilon-squared	0.01 – < 0.08	0.08 – < 0.26	≥ 0.26

Vargha and Delaney's A

```
library(effsize)
VD.A(d = Data$Likert,
     f = Data$Speaker)

Vargha and Delaney A
A estimate: 0.05 (large)

library(rcompanion)
vda(Likert ~ Speaker, data=Data)

VDA
0.05

library(rcompanion)
vda(Likert ~ Speaker, data=Data, ci=TRUE)

VDA lower.ci upper.ci
1 0.05      0      0.162

### Note: Bootstrapped confidence interval may vary.
```

Cliff's delta

```
library(effsize)
cliff.delta(d = Data$Likert,
            f = Data$Speaker)

Cliff's Delta
delta estimate: -0.9 (large)
95 percent confidence interval:
  lower      upper
-0.9801533 -0.5669338

library(rcompanion)
cliffDelta(Likert ~ Speaker, data=Data)

Cliff.delta
-0.9
```

```
library(rcompanion)
cliffDelta(Likert ~ Speaker, data=Data, ci=TRUE)

      cliff.delta lower.ci upper.ci
1      -0.9      -1      -0.67

### Note: Bootstrapped confidence interval may vary.
```

r

```
library(rcompanion)
wilcoxonR(x = Data$Likert,
          g = Data$Speaker)

      r
0.791

library(rcompanion)
wilcoxonR(x = Data$Likert,
          g = Data$Speaker,
          ci = TRUE)

      r lower.ci upper.ci
1 0.791   0.602   0.897

### Note: Bootstrapped confidence interval may vary.
```

tau-b

```
library(DescTools)
KendallTauB(x = Data$Likert,
            y = as.numeric(Data$Speaker))

[1] 0.7397954

library(DescTools)
KendallTauB(x = Data$Likert,
            y = as.numeric(Data$Speaker),
            conf.level = 0.95)

      tau_b      lwr.ci      upr.ci
0.7397954 0.6074611 0.8721298
```

Freeman's *theta*

```
library(rcompanion)
freemanTheta(x = Data$Likert,
             g = Data$Speaker)

      Freeman.theta
0.9

library(rcompanion)
freemanTheta(x = Data$Likert,
             g = Data$Speaker,
             ci = TRUE)
```

	Freeman.theta	lower.ci	upper.ci
1	0.9	0.688	1

Note: Bootstrapped confidence interval may vary.

epsilon-squared

```
library(rcompanion)
```

```
epsilonSquared(x = Data$Likert,  
              g = Data$Speaker)
```

epsilon.squared
0.658

```
library(rcompanion)
```

```
epsilonSquared(x = Data$Likert,  
              g = Data$Speaker,  
              ci = TRUE)
```

	epsilon.squared	lower.ci	upper.ci
1	0.658	0.383	0.842